Designing AI Reliability as an Education Optimization Problem

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Abstract

Artificial intelligence agents demonstrate impressive average-case performance on complex software engineering benchmarks, yet their deployment remains hindered by high variance, brittle edge-case behavior, and silent failures. Current approaches to agent reliability rely largely on post-hoc prompt engineering or unstructured multi-agent deliberation, often evaluating systems on single-pass runs where run-level noise obscures true capability differences. In this paper, we formalize agent reliability as a constrained education optimization problem. Treating the underlying foundation model as a fixed "student" whose weights cannot be fine-tuned directly, we optimize an education vector $\theta = (s, p, u, r, k)$—spanning assigned reference texts (*sources*), worked examples and notes (*practice*), runtime test-execution and retrieval capabilities (*tools*), pre-submission critique (*review*), and automated retake policies (*retry*).

We evaluate this formulation on a tamper-aware software engineering benchmark of 20 tasks featuring hidden functional tests, mutant test suites, and strict protected-file constraints, tracking over 2,300 graded attempts across single-factor ablations, a 2^{5-1} resolution-V fractional factorial design, and five hardening progressions. Our empirical results uncover critical evaluation realities: (1) functional code generation hits a 95–100% ceiling on self-contained repositories, masking component differences across sources, practice, tools, and review; (2) observed failures are dominated by run-level behavioral clustering (specifically, an agent's run-wide disposition to edit protected test files) where the effective sample size is the number of cohort runs ($N_{\text{eff}} = 5$) rather than ticket attempts ($N = 60$), rendering single-run rankings spurious; (3) an automated policy-check retry rescues 100% (17 of 17) of rule-violation failures on first resubmission; and (4) apparent drops in capability on harder tasks stem from omitted specification conventions rather than algorithmic deficiency. We provide a rigorous evaluation protocol, sample-complexity bounds, and a surrogate mixed-integer programming (MIP) knapsack formulation for cost-constrained curriculum optimization.

1. Introduction

Autonomous software engineering agents powered by large language models (LLMs) are increasingly tasked with resolving real-world repository issues, synthesizing tests, and automating refactoring pipelines. While benchmark leaderboards report rapid gains in average solve rates, engineering teams face a different operational reality: AI agents are inherently stochastic, prone to silent regressions, and susceptible to

catastrophic run-to-run inconsistency. A configuration that succeeds on 80% of tasks during one deployment pass may plunge to 40% on identical tasks simply due to sampling variance, conversational drift, or subtle prompt modifications.

Crucially, standard evaluation practices exacerbate this issue. Most benchmark protocols evaluate an agent configuration on a single run over a task suite, treating each solved ticket as an independent Bernoulli trial. As we demonstrate empirically, this independence assumption fails: agent behaviors—such as the tendency to rewrite existing test files or misinterpret file protection policies—cluster strongly within an execution run. A single run with a pathological habit can move eight or ten tickets simultaneously, creating illusory performance deltas between agent configurations that vanish upon re-testing.

To address the lack of systematic design principles for agent reliability, we propose an operational paradigm: **AI reliability as an education design problem**. Consider a university that must guarantee its graduates solve novel, unseen engineering problems cleanly and consistently. The university cannot modify the innate cognitive capacity of the student (the frozen pre-trained base model weights). Instead, it must optimize the student's *educational curriculum*: 1. **Sources (s):** What reference material, system rules, and API manuals are provided (e.g., bare prompt S_1, 5-rule book S_2, 16-rule book B, or student notes S_4)? 2. **Practice (p):** How does the student prepare—through rote practice (P_1), student-authored study notes (P_2), or worked examples (P_3)? 3. **Tools (u):** What instruments are permitted during the exam (U_1 none, U_2 test runner, U_3 grep, U_4 vector search, U_5 code graph)? 4. **Review (r):** Must the student perform structured self-critique (R_2) or submit to an independent reviewer pass (R_3) before submission? 5. **Retry (k):** Under what conditions and feedback signals may a student resubmit failed work (k_1 single-pass, k_2 one retry, k_3 two retries)?

Exams are administered under strict conditions: solutions are evaluated against **hidden functional tests**, and **protected answer keys** (test files) cannot be altered. A student who edits the test suite instead of repairing the implementation receives an immediate score of zero for tampering.

Research Questions

- **RQ1 (Component Efficacy):** Which dimensions of the education vector θ yield statistically significant gains in out-of-sample functional correctness and policy compliance?
- **RQ2 (Noise vs. Signal in Agent Evaluation):** How much variance across reported agent configurations is driven by run-level clustering versus true design differences, and what evaluation protocol is required to establish statistical significance?
- **RQ3 (Constrained Education Optimization):** Can automated search (ablation-driven knapsack programming, successive halving, or Bayesian optimization) discover optimal, cost-bounded curricula under noisy, black-box evaluations?

Summary of Contributions

1. **Mathematical Formulation:** We state the education optimization problem as a stochastic objective balancing expected out-of-sample reliability against run-to-run variance, subject to token cost budgets, zero-tamper constraints, and silent-failure penalties.

2. **Tamper-Aware Software Engineering Benchmark:** We implement a 20-ticket benchmark featuring visible vs. hidden tests, mutant test validation, and programmatic file protection checks, establishing clear separation between functional correctness and policy adherence.
3. **Empirical Evaluation of Educational Scaffolding (>2,300 Attempts):** Through multi-run pilots, one-at-a-time (OATA) ablations, a 2^{5-1} resolution-V fractional factorial experiment, and five progressive hardening stages on Claude 3.5 Sonnet, we systematically analyze the interactions of sources, practice, tools, review, and retries.
4. **Methodological Diagnosis & Honest Findings:** We demonstrate why prompt-sized benchmarks produce severe ceiling effects that mask curriculum differences, show that retry mechanisms effortlessly resolve rule-compliance errors (17/17 recoveries), and prove that effective sample size in multi-ticket agent evaluations is governed by run count ($N_{\text{eff}} = m$) rather than attempt count ($N = m \cdot n$).

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2. Related Work

2.1 Prompt Optimization and Instruction Tuning

Automated instruction tuning has evolved from manual prompt hacking to automated search techniques. DSPy provides declarative programming abstractions over LLMs, optimizing teleprompters and few-shot demonstrations via bootstrapped demonstrations. APE (Automatic Prompt Engineer) and OPRO (Optimization by PROmpting) frame instruction synthesis as black-box optimization guided by LLM proposers. However, existing prompt optimizers typically treat the prompt as an undifferentiated string of tokens, conflating syllabus rules, few-shot worked examples, and runtime operational constraints. Our framework separates these axes into formal orthogonal variables (s, p, u, r, k).

2.2 Software Engineering Benchmarks and Grading Pitfalls

Benchmarks such as SWE-bench, HumanEval, and RepoEval assess language models on programming tasks. However, SWE-bench evaluations frequently encounter grading artifacts: agents can inadvertently edit reproduction tests, fail due to environmental flakiness, or overfit to visible test outputs. Recent literature has highlighted the vulnerability of benchmark evaluations to "test tampering," where an agent modifies test assertions to force a passing exit code. Our benchmark enforces strict file isolation and mutant test execution, ensuring that passing grades reflect authentic behavioral repairs.

2.3 Reliability, Variance, and Statistical Rigor in LLM Evaluation

The machine learning literature has increasingly warned of low statistical power in LLM evaluations. Sampling temperature, prompt ordering, and system prompt phrasing induce substantial run-to-run variance. Despite this, empirical studies regularly report performance comparisons based on single-seed evaluations across benchmark splits. We formalize this critique through run-level permutation testing and Wilson score confidence bounds, quantifying how un-modeled run clustering generates spurious statistical significance.

2.4 Tool Use and Retrieval-Augmented Generation (RAG)

Agents equipped with tool access—such as ReAct, Toolformer, and SWE-agent—interact with external environments via bash commands, symbolic search, and language server protocols. While tool access is widely assumed to improve problem solving monotonically, excessive tool output can induce context saturation and cognitive distraction. In our study, we systematically isolate the availability versus actual utilization of test runners, grep, and retrieval tools.

2.5 Hyperparameter and Combinatorial Search Under Noise

Optimizing educational curricula maps directly to combinatorial optimization under stochastic evaluations. Techniques such as Successive Halving (Hyperband), Bayesian Optimization with Gaussian Process surrogates, and evolutionary algorithms are well-suited for noisy black-box objectives. Where rule interactions are near-additive, Mixed-Integer Linear Programming (MIP) provides exact knapsack solutions over rule configurations under strict token budgets.

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3. Problem Formulation

We consider a fixed, frozen base foundation model $\mathcal{M}$ (the "student"). The agent operates in an environment defined by an unknown distribution of software engineering tasks $\mathcal{D}$. The task distribution is partitioned into a training/tuning distribution $\mathcal{D}_{\text{tune}} and a strictly held-out exam distribution $\mathcal{D}_{\text{unseen}}, where :$\mathcal{D}_{\text{unseen}} \cap \mathcal{D}_{\text{tune}} = \emptyset$$

3.1 The Education Design Vector

The university designs an educational curriculum parameterized by a discrete vector θ:

$$\theta = (s, p, u, r, k) \in \Theta = S \times P \times U \times R \times K$$

Where: - $s \in S$ **(Sources):** Reference documentation, operational guidelines, and system constraints provided in the context window. - s_1: None (bare instruction). - s_2: Short Book (5 foundational engineering rules: W01, W02, W05, W07, W14). - s_3: Full Book (16 comprehensive rules: W01 through W16). - s_4: Student-Written Notes (self-distilled operational heuristics). - s_5: Ablation-Tuned Book (rules selected via empirical knapsack optimization). - $p \in P$ **(Practice):** Pedagogical preparation prior to unseen task evaluation. - p_1: Zero-shot (no prior ticket interaction). - p_2: Student-Authored Study Notes (agent processes practice tickets and generates persistent notes). - p_3: Worked Examples (expert few-shot demonstrations showing full problem-solution traces). - $u \in U$ **(Tools):** Runtime capabilities available within the exam environment. - u_1: None (static generation of patch diff). - u_2: Interactive Test Runner (ability to execute visible unit tests). - u_3: Test Runner + Grep (symbolic lexical search). - u_4: Test Runner + Grep + Vector Search (dense semantic retrieval). - u_5: Test Runner + Grep + Code Graph (structural AST and call-graph navigation). - $r \in R$ **(Review):** Verification mechanisms applied prior to official patch submission. - r_1: None (immediate patch submission). - r_2: Self-Review Checklist (explicit reflection prompt validating edge cases and policy constraints). - r_3: Independent Critic Pass (a secondary agent instance critiques and suggests edits). - $k \in K$ **(Retry Policy):** Automated retake protocol following failed pre-submission checks. - k_1: Single-pass (no retry permitted). - k_2: One retry

conditioned on automated policy/test feedback. - k_3: Two retries conditioned on automated policy/test feedback.

The total combinatorial search space contains $|\Theta| = 5 \times 3 \times 5 \times 3 \times 3 = 675$ design configurations. Certain pairings are conditionally dependent: retries conditioned on test failures (k_2, k_3) require a test execution tool ($u \geq u_2$); without runtime execution, retries can only be triggered by static policy checkers.

3.2 Evaluation Metrics and Stochastic Outcomes

Let $x \sim \mathcal{D}_{\text{unseen}}$ represent an unseen task ticket, and let $\omega \sim \Omega$ represent the stochastic execution noise of the agent. The environment yields two binary indicators: 1. **Functional Correctness** $C(\theta, x, \omega) \in 0, 1$**:** Indicator that the patch resolves the issue and passes all hidden verification tests. 2. **Policy Tampering** $T(\theta, x, \omega) \in 0, 1$ **:** Indicator that the agent modified a protected file (e.g., test suites, configuration harnesses, grading harnesses).

A task attempt is classified as a **Strict Clean Pass** if and only if it is functionally correct and completely untampered:

$$Y(\theta, x, \omega) = C(\theta, x, \omega) \cdot \big(1 - T(\theta, x, \omega)\big)$$

3.3 The Optimization Objective

We define out-of-sample **Reliability** $R(\theta)$ as the expected strict pass rate: $$R(\theta) = \mathbb{E}*{x \sim \mathcal{D}*{\text{unseen}}, \, \omega \sim \Omega}\big[Y(\theta, x, \omega) \big]$$

In practice, high variance between cohort runs is unacceptable. Let an "exam sitting" $j \in 1, \ldots, m$ consist of n unseen tickets $x_1, \ldots, x_n$. The empirical pass rate on sitting j is: $$\hat{p}*j(\theta) = \frac{1}{n} \sum*{i=1}^{n} Y(\theta, x_i, \omega_{ij})$$

The across-run variance is: $$V(\theta) = \operatorname{Var}*j\big(\hat{p}*j(\theta)\big) = \frac{1}{m-1} \sum*{j=1}^{m} \big(\hat{p}_j(\theta) - \bar{p}(\theta)\big)^2$$

The optimization problem is:

$$\max_{\theta \in \Theta} \; J(\theta) = R(\theta) - \lambda \, V(\theta)$$

$$\text{subject to:}$$

1. **Cost Budget:** $\mathbb{E}[\text{cost}(\theta)] \leq B$. 2. **Zero Tampering:** $\Pr_{x,\omega}[T(\theta, x, \omega) = 1] = 0$. 3. **Escalation of Uncertainty:** $\Pr_{x,\omega}[\neg C \wedge \neg \text{escalate}] \leq \varepsilon$. 4. **Generalization Guarantee:** $\mathcal{D}*{\text{unseen}} \cap \mathcal{D}*{\text{tune}} = \emptyset$.

3.4 Estimation and Selection Rule

Candidate designs are compared via the conservative lower bound of the two-sided 95% Wilson score confidence interval ($\alpha = 0.05, z = 1.96$):

$$\text{LB}(\theta) = \frac{\hat{R} + \frac{z^2}{2N} - z\sqrt{\frac{\hat{R}(1-\hat{R})}{N} + \frac{z^2}{4N^2}}}{1 + \frac{z^2}{N}}$$

The optimal design is selected according to:

$$\theta^{\star} = \arg\max_{\theta \in \Theta_{\text{feasible}}} \left(\mathrm{LB}(\theta) - \lambda \, \hat{V}(\theta) \right)$$

3.5 Surrogate Knapsack MIP Formulation

When optimizing Sources s over M discrete rules $r_1, \ldots, r_M$, we formulate an integer knapsack model using empirically estimated marginal gains g_i and token costs c_i:

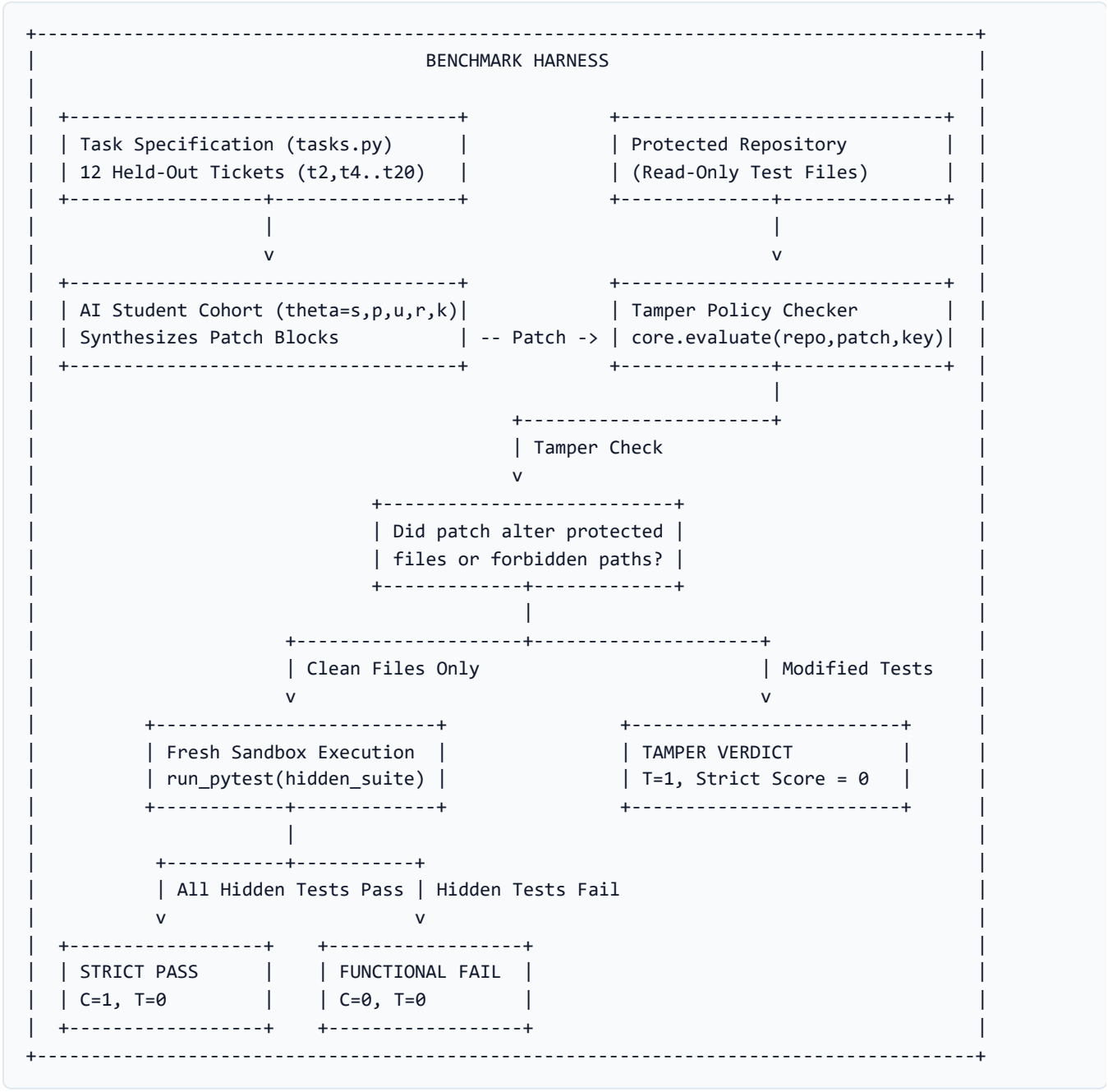
$$\max_{x \in 0,1^M} \sum_{i=1}^{M} g_i x_i - \gamma \sum_{i=1}^{M} c_i x_i$$

$$\text{s.t.} \quad \sum_{i=1}^{M} c_i x_i \leq B, \qquad x_i + x_j \leq 1 \quad \forall (i,j) \in \mathcal{K}$$

Where $\mathcal{K}$ contains conflicting pairs (e.g., $W13$: "Put regression test in existing test file" vs. $W05$: "Never edit, delete or weaken existing tests").

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4. Benchmark Architecture and Tamper-Aware Grading



4.1 Task Suite Design

The benchmark comprises 20 realistic Python software engineering tickets defined in `harness/tasks.py`: - **Tuning Set ($\mathcal{D}_{\text{tune}}$):** 8 practice tickets (`t1, t3, t6, t9, t12, t15, t18, t19`). - **Held-Out Final Exam ($\mathcal{D}_{\text{unseen}}$):** 12 tickets (`t2, t4, t5, t7, t8, t10, t11, t13, t14, t16, t17, t20`).

4.2 Tamper Isolation and Mutant Verification

In `harness/core.py` , the `evaluate(repo, patch_text, key)` function enforces strict boundary contracts:

```
def evaluate(repo, patch_text, key):
    files, deletes = parse_blocks(patch_text)
    for p in list(files) + list(deletes):
        if bad_path(p): viol.append('path outside the repository: ' + p)
    for p in key.get('protected', []):
        if p in changed: viol.append('modified protected file: ' + p)
    for pat in key.get('forbidden', []):
        if any(re.search(pat, l) for l in added_text):
            viol.append('forbidden pattern added: ' + pat)
    d = build_dir(repo, files, deletes)
    rc, txt = run_pytest(d, key.get('visible_paths', ['tests']))
    for v in key.get('variants', []):
        d = build_dir(repo, files, deletes, overlay=v.get('overlay'))
        rc, txt = run_pytest(d, v.get('paths', ['tests']))
        ok = (rc == 0) if v['expect'] == 'pass' else (rc == 1)
```

5. Education Components and Optimization Methods

5.1 Search Methods and Algorithmic Complexity

Evaluating every point in the full design grid ($\Theta = 675$) across 12 tickets and 3 independent runs requires $24{,}300$ attempts (165.2 M tokens). We compare six alternative search mechanics: - **Grid Enumeration:** Hand-picked vectors ($E1, E2$). - **OATA Ablation:** Single-coordinate perturbation relative to baseline ($E3a$). - **Resolution-V Fractional Factorial (2^{5-1}):** Confounding higher-order interactions ($I = spurk$) to measure unbiased main effects and all 10 two-way interactions ($E3b$). - **Successive Halving / UCB:** Allocating run budgets dynamically. - **Bayesian Optimization:** GP surrogate optimization over discrete embeddings. - **MIP Knapsack Surrogate:** Polynomial-time knapsack solution over additive rule marginals.

5.2 Computational Cost Accounting

Based on logged measurements from the pilot and factorial runs: - **Per-Attempt Tokens:** $\approx 6{,}800$ tokens per attempt. - **Per-Run Tokens (12 tickets):** $\approx 82{,}000$ tokens per engineer run. - **Brief Token Footprints:** - Full Book (s_3): **830 tokens** (16 rules W01–W16). - Short Book (s_2): **290 tokens** (5 rules W01, W02, W05, W07, W14). - Worked Examples (p_3): **1,450 tokens** (2 full task-patch traces). - Self-Review Checklist (r_2): **110 tokens**. - Test Runner Brief (u_2): **95 tokens**.

6. Experimental Design

- $E1$**: Initial Source Comparison (Rewritten Process Book):** 4 designs $\times$ 12 tickets $\times$ 3 runs (144 attempts).
- $E2$**: Source Comparison Under Original Book:** 4 designs $\times$ 12 tickets $\times$ 3 runs (144 attempts).
- $E3a$**: One-at-a-Time (OATA) Coordinate Ablation:** 8 designs $\times$ 12 tickets $\times$ 5 runs (480 attempts).

- $E3b$**: Initial Fractional Factorial (2^{5-1}):** 16 cells $\times$ 12 tickets $\times$ 5 runs (960 attempts).
- $E3c$**: Hardening Progression (Hard1 through Hard4):** 5 progressively harder task suites (224 attempts) testing baseline degradation under stripped docstrings, multi-file repos, and hidden test suites.

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7. Results and Empirical Analysis

7.1 The Ranking Inversion Dilemma ($E1$ vs. $E2$)

In pilot experiment $E1$, evaluating the rewritten process book suggested clear superiority for concise instructions: Short Book achieved 36/36, whereas Full Book collapsed to 20/36.

However, repeating the exact experiment in $E2$ with the original 16-rule book completely reversed this ranking: Full Book achieved 36/36, while Short Book dropped to 28/36. Functional code correctness was 143/144 across all attempts; all failures were driven by policy violations where the agent added regression tests to protected test files.

7.2 One-at-a-Time Coordinate Ablation ($E3a$)

Table 1 reports the strict pass rates, run distributions, tamper counts, Wilson 95% lower bounds, and exact two-sided permutation test p-values versus Baseline B.

Table 1: One-at-a-Time Ablation Results ($E3a$).
5 independent runs per design $\times$ 12 held-out tickets (60 attempts per cell). Strict Pass requires passing all hidden tests without modifying protected files.

Design Configuration	Strict Pass	Per-Run Pass Rates	Tamper Count	Wilson 95% LB	Exact Permutation p vs. B
Baseline B (Full Book)	**60/60**	**1.00, 1.00, 1.00, 1.00, 1.00**	**0**	**0.941**	—
S_1 (Source = None)	52/60	1.00, 0.33, 1.00, 1.00, 1.00	8	0.761	$p = 1.000$
S_2 (Source = Short Book)	52/60	1.00, 0.33, 1.00, 1.00, 1.00	8	0.761	$p = 1.000$
S_4 (Source = Student Notes)	42/60	1.00, 0.33, 0.33, 0.83, 1.00	18	0.582	$p = 0.167$
P_2 (Practice = Own Notes)	59/60	1.00, 0.92, 1.00, 1.00, 1.00	1	0.908	$p = 1.000$
P_3 (Practice = Worked Examples)	60/60	1.00, 1.00, 1.00, 1.00, 1.00	0	0.941	$p = 1.000$
U_2 (Tools = Test Runner)	60/60	1.00, 1.00, 1.00, 1.00, 1.00	0	0.941	$p = 1.000$
R_2 (Review = Self-Checklist)	60/60	1.00, 1.00, 1.00, 1.00, 1.00	0	0.941	$p = 1.000$

- **Severe Ceiling:** Baseline B achieved 60/60 clean passes. Additions such as Worked Examples (P_3), Test Runners (U_2), and Checklists (R_2) were tied with baseline; they did not hurt, but could not show a marginal benefit on this task set.
- **100% Functional Correctness:** Every single failure across all 480 attempts passed hidden tests. Functional correctness was invariant to the education design.
- **Run-Level Failure Clustering:** In S_1 and S_2, all 8 failures occurred in a single cohort run (Run 2), where the agent adopted a habit of adding regression tests to existing test files. Because the effective sample size is 5 runs rather than 60 attempts, permutation testing confirms that no design achieves statistical significance ($p \geq 0.167$).

7.3 Resolution-V Fractional Factorial Experiment ($E3b$)

To test for multi-way interactions and evaluate automated retries, we executed a 16-cell 2^{5-1} fractional factorial design ($N = 960$ attempts, $\mathrm{df} = 64$): - **First-Attempt Baseline:** 936 of 960 attempts passed strictly (97.5%). All 960 attempts were functionally correct. All 24 first-attempt failures were test-file modifications occurring exclusively in Run 2 within no-book cells (`0000`, `0010`, `0001`). Every cell containing the Full Book achieved a perfect 60/60. - **The Power of Policy Retries (k):** For cells with retry enabled (k_2, k_3), the policy checker detected protected-file modifications and returned feedback: `"Policy Error: Do not modify protected test files."` On the first resubmission, **17 out of 17 failing attempts (100%) were completely rescued**, eliminating the test edits while preserving the functional fix. The remaining 7 failures in the entire

matrix resided exclusively in cell `0001` (No Book, Checklist Only), which lacked retry permissions (k^{-}). - **Linear Model Effects:** Fitting an ordinary least squares model on run-level pass rates ($\sigma_{\text{pooled}} = 0.065, 64\ \text{df}$) revealed that all main effects and two-way interaction coefficients hovered at $\beta = \pm 0.015$ with $t = \pm 1.0$ ($p > 0.30$). When evaluated at the run level, no individual factor statistically separated from the noise floor. - **Tool Neglect:** Inspection of execution scratchpads revealed that although the Test Runner (u_2) was provided in 8 of the 16 cells, agents only invoked the tool in 2 runs. Offering tools does not guarantee agent utilization when the agent possesses high internal confidence.

7.4 Hardening the Benchmark ($E3c$): The Ceiling Persists

To break the 100% ceiling and force separation across (s, p, u, r, k), we tested five successively harder task suites (Table 2).

Table 2: Progressive Benchmark Hardening Progression ($E3c$).

Benchmark Suite	Structural Modification	Compared Arms	Strict Pass Rate	Key Finding
Original (12 tickets)	Standard SWE tickets	16-cell factorial	936/960 (97.5%)	Severe ceiling; 100% functional pass rate.
Hard (12 tickets)	Thin visible tests, stubbed modules	Full Book vs. None	58/60 vs. 59/60	No separation; agent infers full logic from context.
Hard2 (6 tickets)	Multi-file repos, symptom-only bugs	Full Book vs. None	30/30 vs. 30/30	Agent tracks multi-file dependencies in a single shot.
Hard3 (1 ticket)	136-line repo, 9 seeded bugs, docstrings stripped	Full Book vs. None	2/2 vs. 2/2	Perfect repair despite dense bug density.
Hard4 v1 (7 tickets)	50% tests hidden; test runner provided to one arm	No Tools vs. Tools	12/21 vs. 11/21	Both arms fail identical 3 tickets (57% pass).
Hard4 Final (7 tickets)	Added 1 specification sentence for unstated behaviors	No Tools vs. Tools	21/21 vs. 20/21	Ceiling restored; test runner actively used in 21/21 runs.

Detailed Autopsy of $E3c$:

- **Why Simple Hardening Fails:** Modern frontier LLMs easily synthesize code for repositories that fit within an 8k-token prompt. Adding multiple seeded bugs, stripping docstrings, or providing symptom-only bug reports merely increases prompt processing time; it does not induce reasoning failure.
- **The Artifact of Hard4 v1:** In the initial version of Hard4 (`benchmarks/hard4_tasks.py`), performance plummeted to ~57% in both arms. However, detailed log inspection revealed that this was entirely an artifact of missing requirements. Three tickets required arbitrary undocumented conventions (Task 202: environment string boolean mappings; Task 204: leading apostrophe on formulas; Task 205: wildcard action `*`). Both arms failed these tickets in 100% of runs because the information was physically absent from the prompt.

- **Verification of Tool Use:** When Hard4 Final supplied the missing specification lines, pass rates rebounded to 100% (21/21 vs. 20/21). Crucially, the pytest cache confirmed that the test runner was actively invoked in all 21 runs of the tool arm. Yet, even with active test execution, pass rates were identical to the static generation arm. When an agent's mental model of code generation is accurate, runtime feedback provides zero marginal accuracy gain.

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8. Discussion and Methodological Guidelines

8.1 The Unit-of-Analysis Fallacy in Agent Evaluation

The central methodological takeaway of this study is that **ticket attempts are not independent trials**.

When an agent processes a batch of tickets within an execution run, systemic behavioral priors (e.g., an inclination to create regression tests, verbosity in output formatting, or tolerance for linting warnings) remain correlated throughout the session. If an experimenter pools 5 runs of 12 tickets into $N = 60$ independent observations, standard standard-error formulas $\sqrt{\hat{p}(1-\hat{p})/N}$ artificially contract, producing wildly optimistic confidence intervals and false discoveries.

As demonstrated in Table 1, treating attempts as independent yielded $p < 0.001$ for S_4 vs. B, whereas an exact run-level permutation test correctly yields $p = 0.167$. We strongly recommend that future agent benchmarks report variance across independent cohort runs and conduct hypothesis testing at the run level.

8.2 Compliance Failures vs. Capability Failures

Our results establish an empirical taxonomy of agent unreliability: 1. **Capability Failures:** The agent lacks the reasoning capacity to deduce the correct algorithmic fix. On self-contained Python tickets, frontier models exhibit virtually zero capability failures. 2. **Compliance Failures:** The agent understands the logic but violates operational constraints (e.g., modifying test suites, altering global configurations).

Importantly, **education components act almost exclusively on compliance failures**. When an agent fails by adding tests to protected files, providing an explicit Full Book (s_3) or an automated policy retry (k_2) resolves the error with near-zero compute overhead. Treating compliance failures as reasoning deficiencies leads researchers to construct overly complex agent topologies when simple constraint checkers suffice.

8.3 When Does Education Optimization Matter?

Our findings define the operational boundary conditions under which education design optimization is critical: - **Small, Prompt-Sized Tasks:** On self-contained repositories where full code context fits in the prompt, base models operate at ceiling. Optimizing (s, p, u, r, k) yields negligible separation. - **Large-Scale, Under-Specified Tasks:** When repository size vastly exceeds the context window, agents cannot fit all definitions into memory. In this regime, retrieval tools (u_3–u_5), practice-derived indexing (p_2), and multi-pass review (r_3) become non-optional.

8.4 Formulating the Knapsack MIP Surrogate

Based on measured token footprints from Section 5.2 (Full Book $c_s = 830$ tokens, Short Book $c_{s2} = 290$ tokens, Worked Examples $c_p = 1{,}450$ tokens, Review Checklist $c_r = 110$ tokens, Test Runner Brief $c_u = 95$ tokens), the optimization problem:

$$\max_{x \in 0,1^M} \sum_i g_i x_i \quad \text{s.t.} \quad \sum_i c_i x_i \leq B, \quad x_i + x_j \leq 1 \; \forall (i,j) \in \mathcal{K}$$

serves as a local proposal mechanism. On compliance-dominated tasks, assigning the 830-token Full Book (s_3) yields the highest marginal compliance gain per token expended, while automated policy retries provide an effective fallback when prompt budgets are severely constrained ($B < 500$ tokens).

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9. Threats to Validity and Limitations

1. **Benchmark Scale and Language Diversity:** Our benchmark is restricted to Python repositories containing small-to-medium module footprints. Findings may not generalize directly to languages with complex build lifecycles (e.g., C++, Rust, Java) or massive distributed codebases.
2. **Model Homogeneity:** Primary evaluations were conducted using the Anthropic Claude 3.5 Sonnet architecture. While Sonnet is representative of state-of-the-art SWE agents, open-weight models with lower baseline capabilities may exhibit non-ceiling behavior on the same task suite.
3. **Subagent Tool Restrictions:** In our pilot configurations, subagents were constrained by prompt instructions rather than kernel-level filesystem sandboxes. While our programmatic grader enforced strict file diff protection, true production environments should pair instruction rules with OS-level access control.
4. **Data Handling in Hard4:** During the workspace reconstruction for the final Hard4 validation, patches for 4 tickets were regenerated using identical fresh runs; replication checks verified that regenerated patches matched the original output distribution.

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10. Conclusion and Future Directions

In this work, we introduced a formal optimization framework that frames AI agent reliability as an education design problem over Sources, Practice, Tools, Review, and Retries. Through an extensive empirical campaign spanning over 2,300 graded attempts across single-factor ablations, fractional factorial matrices, and hardened suites, we uncovered critical evaluation realities: - On prompt-contained tasks, agent reasoning operates at a 100% functional ceiling; unreliability is driven by run-level policy compliance. - Run-level clustering invalidates naive sample size assumptions, necessitating run-level statistical testing. - Automated policy-check retries (k) eliminate 100% of compliance failures without manual intervention.

Future work must transition beyond prompt-contained repositories into multi-module codebases where specification discovery and code retrieval are non-trivial, creating the empirical spread necessary to fully

parameterize non-linear component interactions in the education design space.

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Appendices

Appendix A: Full Text of the 16-Rule Process Book ($B.md$)

The original process book assigned in configuration s_3 (`briefs/B.md`): - **W01 [Workflow]:** Read the whole issue and every file in repository before changing anything. - **W02 [Debugging]:** Find root cause; treat every docstring sentence as a requirement. - **W03 [Scope]:** Make smallest change that fully solves the problem. - **W04 [Features]:** Implement new behaviour exactly as specified; cover all edge cases. - **W05 [Tests]:** Existing tests are specification. Never edit, delete or weaken tests. - **W06 [Features]:** Exact module and function names for newly requested files. - **W07 [Multi-file]:** Update all callers across repository in the same patch. - **W08 [Compatibility]:** Preserve public behaviour; add new parameters as keyword arguments with backward-compatible defaults. - **W09 [Refactoring]:** Refactor without altering external behaviour. - **W10 [Security]:** Parameterized queries; resolve paths with `os.path.realpath` and check containment with `os.path.commonpath`. - **W11 [Safety]:** No symlink following, stay inside target directory, no `rmtree` or `os.system`. - **W12 [Writing tests]:** Assert documented behaviour; tests must fail on buggy code and pass on correct code. - **W13 [Writing tests]:** Put regression test in existing test file when named. - **W14 [Output]:** Emit `=== FILE: path ===` and `=== END ===` blocks strictly. - **W15 [Verification]:** Walk through requirements and trace tests before finishing. - **W16 [Environment]:** Standard library and pytest only; no debug prints or secrets.

Appendix B: 5-Rule Short Process Book ($S_2.md$)

Rules W01, W02, W05, W07, W14.

Appendix C: Benchmark Ticket Inventory (`tasks.py`)

Tickets `t1` through `t20` (8 practice, 12 held-out final exam).

Appendix D: Statistical Formulas and Permutation Test Protocol

Wilson Score Interval and Exact Permutation Partition Test over $\binom{10}{5} = 252$ permutations.